HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026 · PHASE 1

GlycoTwin

A type 2 diabetes digital twin that fuses the health record with a glucose sensor and warns two hours before glucose leaves the safe range

---

Team Snack Overflow · RV College of Engineering, Bengaluru

Yash Saraogi (lead) · Tallam Sri Sai Subramanyam · Yash Gautam

github.com/chxmq/SnackOverflow_RVCE

THE PROBLEM

Between two clinic visits a doctor sees one HbA1c; the patient lives through 8,640 glucose readings

101 million Indians live with diabetes and 136 million with prediabetes (ICMR-INDIAB, Lancet Diabetes & Endocrinology, 2023).

HbA1c is a three-month average. It cannot say when glucose went high, or how low it fell at 3 am.

A 15-minute CGM records 96 readings a day: 8,640 per quarter. No clinician can read that raw.

Hypoglycaemia on insulin or sulfonylureas is the acute danger; repeated post-meal spikes are the slow one.

What is missing

A model of this one patient that:

1. reads the record AND the live stream
2. learns how this person responds to meals
3. says, two hours early, which way glucose is about to leave range
4. shows a doctor why it thinks so

That is our digital twin, scoped to one condition: type 2 diabetes.

HEALTHCARE USE CASE

Use case: a diabetes team watches a panel of CGM-wearing patients and acts before the excursion, not after

	What the twin does	What the clinician can do
Who	Endocrinologist, diabetes educator, ward nurse	Inpatient ward or remote monitoring
Input	Record at admission + live CGM + meal and dose logs	No new hardware beyond a CGM
Hyper alert	Rise above 180 mg/dL within 2 h	Check meal, correction dose, walk
Hypo alert	Fall below 70 mg/dL within 2 h	Snack, hold sulfonylurea / insulin
Twin view	Personal post-meal rise, usual meal times	Tailor dose and diet advice
Explanation	Top 5 drivers per alert	Doctor can overrule a wrong reason

The twin does not dose or prescribe. It is decision support: it ranks who needs attention in the next two hours and explains why.

DATA AND SANDBOX RULES

We used real, anonymised patient data: 100 people with type 2 diabetes, 1170 patient-days of CGM

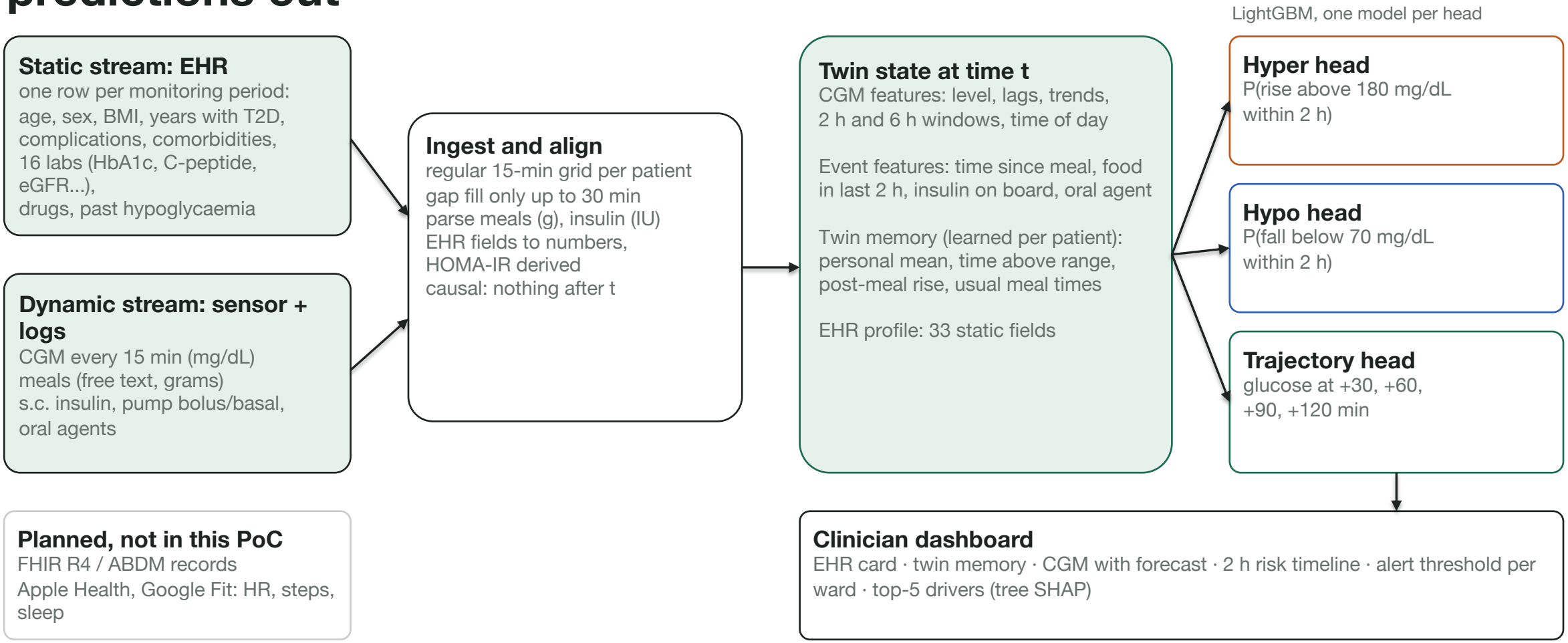
	Size	Detail
Dataset		ShanghaiT2DM (Zhao et al., Scientific Data 10:35, 2023), CC BY 4.0
Patients / periods	100 / 109	Some patients were monitored in more than one period
CGM points	112,287	15-min interval, 3 to 14 days per period
Static (EHR)	33 fields	Age, sex, BMI, smoking, diabetes years, complications, comorbidities, drugs
Labs	16 tests	FPG, 2 h PPG, C-peptide, insulin, HbA1c, glycated albumin, lipids, eGFR, uric acid
Dynamic logs	4 kinds	Meals with grams, s.c. insulin, pump bolus and basal, oral agents

Why not only synthetic data: a model trained on our own simulator would learn our simulator. Real CGM keeps us honest.

Sandbox rules: the data is public, de-identified and openly licensed. No identifiable patient data is used anywhere (DPDP Act and HIPAA).

SYSTEM DESIGN

Architecture: two streams in, one twin state every 15 minutes, three predictions out



THE TWIN

What makes it a twin and not just a forecaster: it keeps a memory of this patient and updates it as data arrives

Twin memory item	How it is learned	Available
Personal mean glucose	Running mean of every CGM reading so far	starts after 2 h
Personal time above range	Share of readings ≥ 180 mg/dL so far	starts after 2 h
Personal post-meal rise	Mean peak rise in the 2 h after each past meal	after first meal + 2 h
Usual meal in next 2 h	Share of past days with a meal logged in this clock window	from day 2
EHR profile	HbA1c, C-peptide, HOMA-IR, drugs, complications	fixed per period

Each item is computed only from data before time t. A meal counts toward the post-meal rise only once its 2-hour window has fully passed, so the label never leaks into the feature.

The twin replays forward: drag the dashboard cursor and the memory grows exactly as it would have at the bedside.

METHOD

Evaluation protocol: every number in this deck is on patients the model never saw

Targets

Hyper: glucose now < 180, and any reading ≥ 180 in the next 2 h.

Hypo: glucose now ≥ 70, and any reading < 70 in the next 2 h.

Splits

5-fold GroupKFold by patient. A patient's periods never straddle train and test.

Alert threshold

Fixed at p ≥ 0.5 before looking at any test fold. Not tuned.

Uncertainty

95% confidence intervals by bootstrapping patients (500 resamples), not windows, because windows from one patient are correlated.

Baselines

Trend rule: current glucose plus twice the last 30-min change, what a nurse would eyeball.

Ablations: EHR only, CGM only, CGM + events, + twin memory, + EHR.

Windows

Hyper: 90,105 windows, 15% positive. Hypo: 109,020 windows, 2.8% positive.

RESULTS

On patients it never saw, the twin flags highs at AUROC 0.878 and lows at 0.907, well above a nurse's trend rule

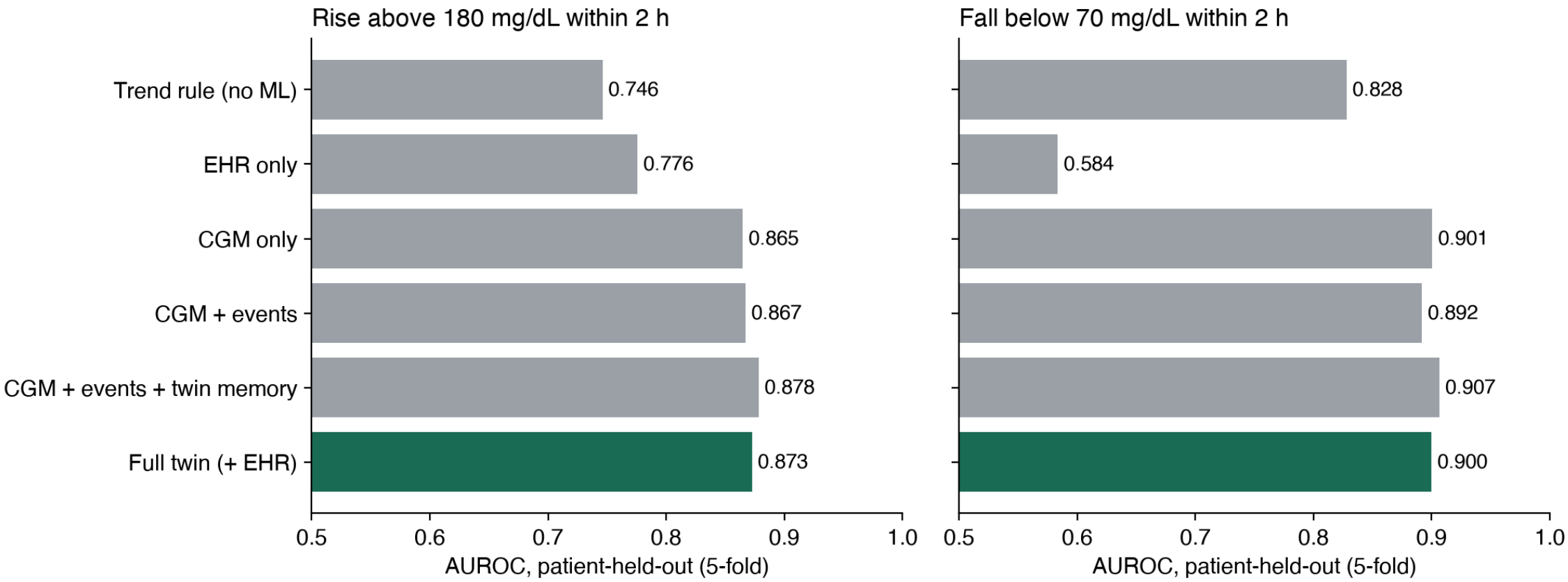
Model (5-fold, patient-held-out)	Hyper AUROC	Hyper AUPRC	Hypo AUROC	Hypo AUPRC
Trend rule (no ML)	0.746	0.388	0.828	0.182
EHR only	0.776	0.381	0.584	0.042
CGM only	0.865	0.570	0.901	0.311
CGM + events	0.867	0.581	0.892	0.274
Personal twin: CGM + events + twin memory	**0.878**	**0.603**	**0.907**	**0.296**
Personal twin + 9 core EHR fields	0.875	0.601	0.898	0.270
Full twin: + all 33 EHR fields (dashboard)	0.873	0.593	0.900	0.270

Full twin minus trend rule, AUROC, 95% CI over patients: highs [+0.107, +0.144], lows [+0.048, +0.100].

Chance level for AUPRC is the prevalence: 0.152 for highs, 0.028 for lows.

ABLATION

The timing comes from the live sensor, the twin's memory of the patient adds to it, and the record alone cannot time an event



WHAT FUSION WITH THE RECORD BOUGHT US

Honest finding: once the twin has a few hours of glucose, the static record adds no measurable accuracy

AUROC gain over CGM only, 95% CI	Highs	Lows	Day-1 highs	Day-1 lows
Personal twin (memory, no EHR)	[+0.007, +0.020]	[-0.004, +0.015]	0.808	0.883
+ 9 core EHR fields	[+0.002, +0.018]	[-0.015, +0.009]	0.813	0.888
+ all 33 EHR fields	[+0.001, +0.015]	[-0.013, +0.010]	0.807	0.866
CGM only (reference)	0	0	0.815	0.907

Highs: the memory gain excludes zero. Lows: every interval includes zero, so we claim no gain there.

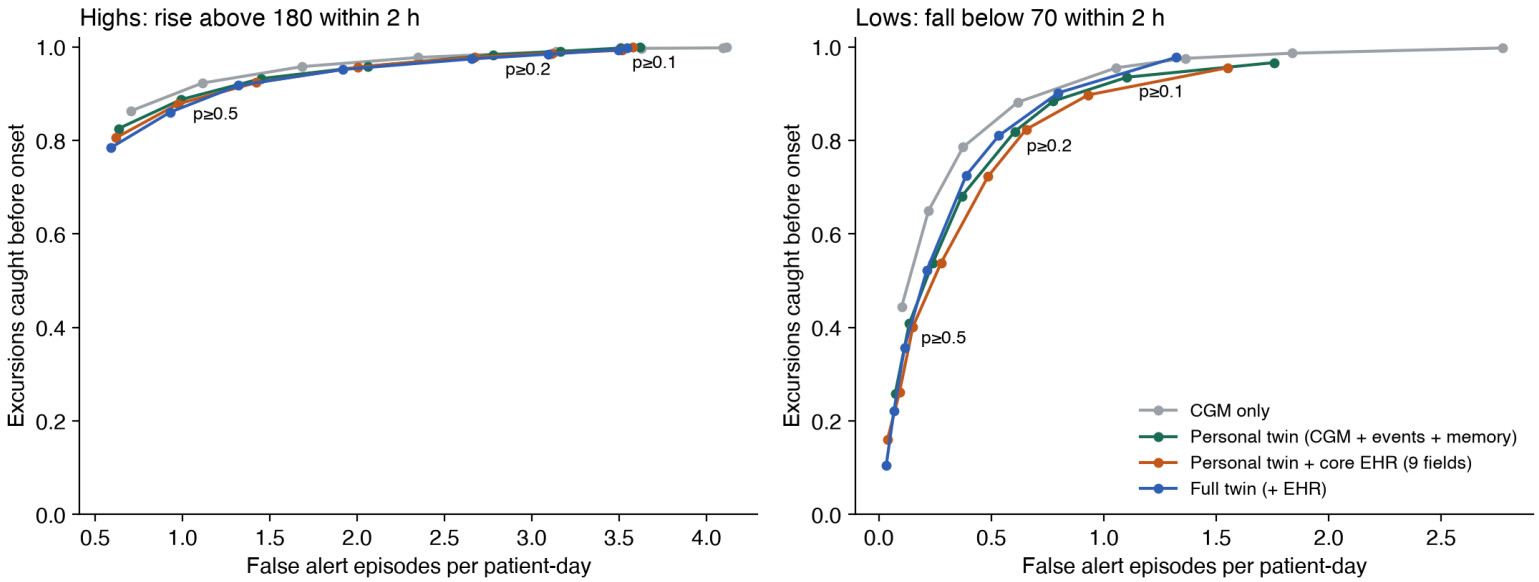
Why the record adds nothing here: the twin's learned mean glucose and time above range are a current, personal version of what HbA1c summarises over three months. With 100 patients, 33 static fields mostly let trees memorise who is who.

The record still carries population risk on its own (EHR-only AUROC 0.776 for highs), and the clinician needs it beside every alert. In the dashboard it is context, not the source of the timing.

Day 1 = the first calendar day of monitoring, before the memory has filled. Even there the record did not add accuracy in this cohort.

ALERT TRADE-OFF

What a ward would see: at p ≥ 0.5 the twin catches 89% of highs before onset; lows are rarer and need a lower threshold

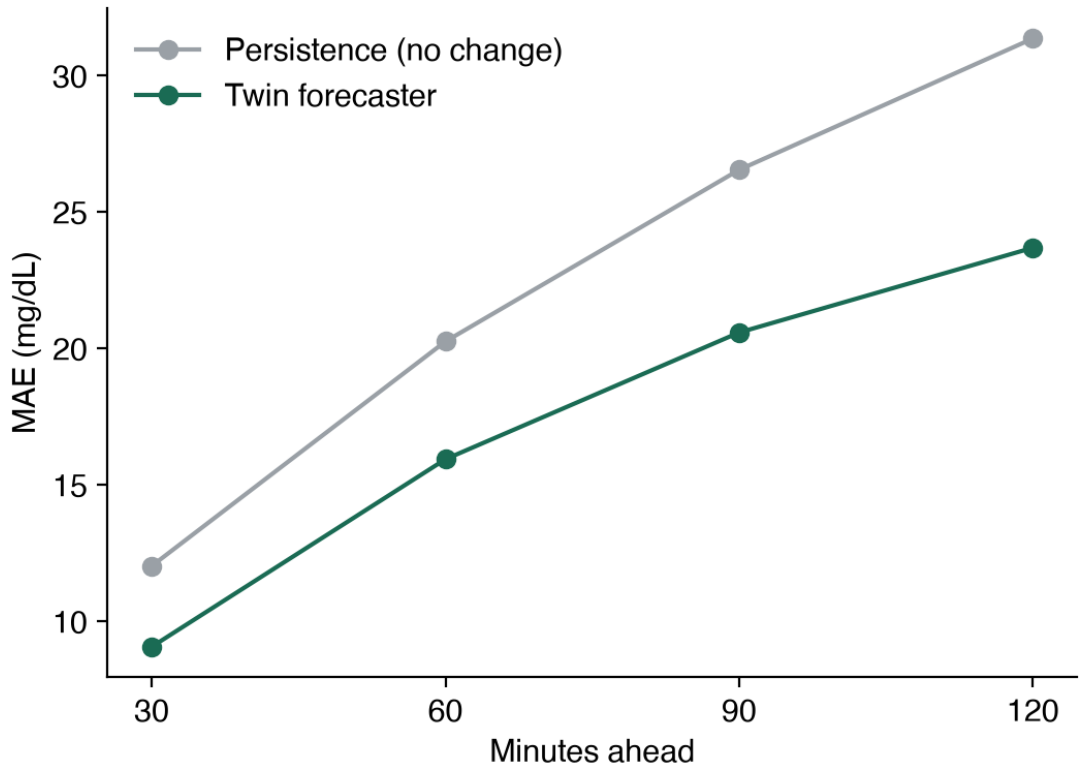


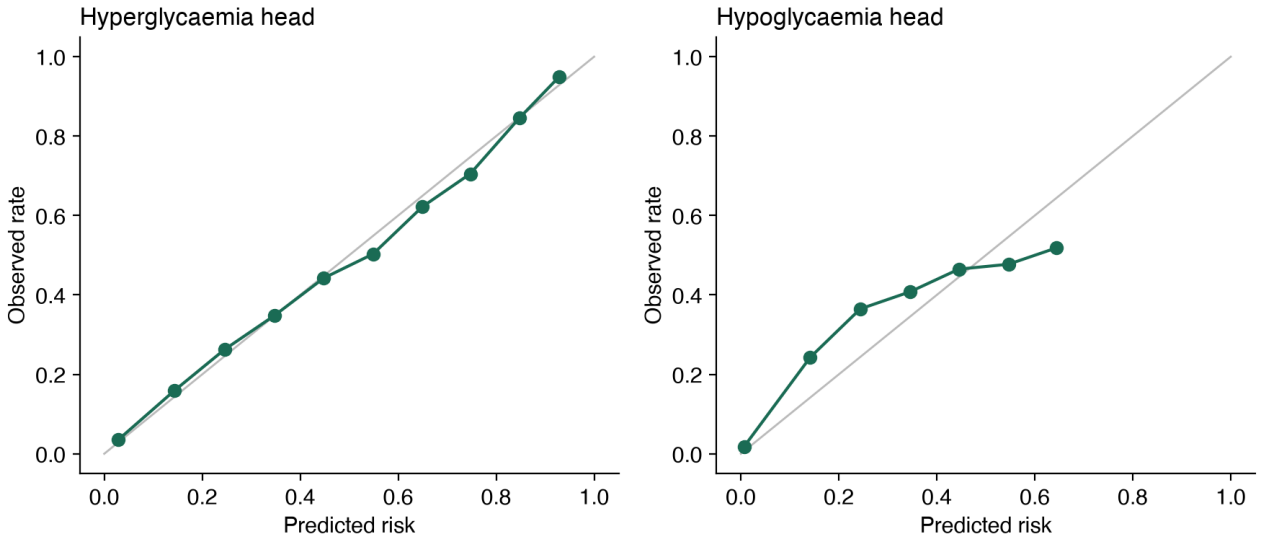
Personal twin	Caught	Lead min	False/day
Highs 0.5	89%	60	0.99
Highs 0.3	96%	90	2.06
Lows 0.5	41%	30	0.14
Lows 0.2	82%	60	0.60
Lows 0.1	94%	75	1.10

Excursions: 1,991 highs, 448 lows in held-out data. Caught = an alert in the 2 h before the crossing. False = an alert run with no crossing in the next 2 h.

0.5 was fixed in advance. The other rows are read off the same curve to show the cost, not tuned results.

The twin also forecasts the trajectory: 15.9 mg/dL mean error at 60 min versus 20.2 for 'no change'



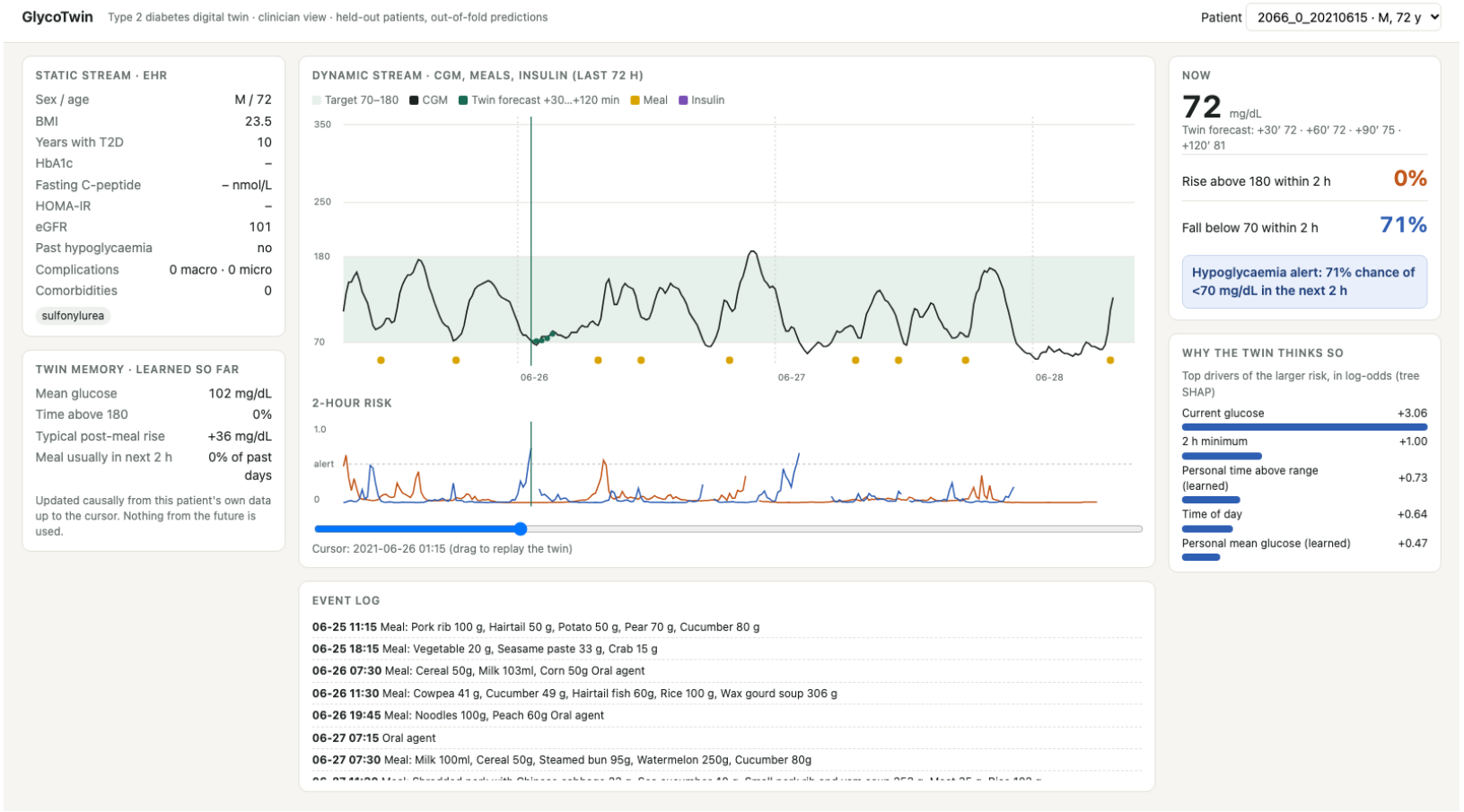


Calibration (right): predicted risk against the observed rate in held-out patients. Points near the diagonal mean a 30% alert really means about 30%.

Per-patient AUROC for the hyper head: median 0.87 (IQR 0.82 to 0.90) across 98 patients with events.

CLINICIAN DASHBOARD

The doctor's view: one screen per virtual patient, replayable, with the reason for every alert



Live: chxmq.github.io/SnackOverflow_RVCE · every number shown is an out-of-fold prediction for a held-out patient

EXPLAINABILITY

What drives the twin: glucose now and its trend, the clock, and what the twin has learned about this patient

Hyper head: feature	mean |SHAP|
Time of day	1.163
Current glucose	0.453
Personal mean glucose (learned)	0.236
15-min trend	0.191
60-min trend	0.148
Time since last meal	0.147
Personal post-meal rise (learned)	0.145
Personal time above range (learned)	0.120

Hypo head: feature	mean |SHAP|
Current glucose	0.472
Personal mean glucose (learned)	0.385
Personal variability (learned)	0.303
2 h minimum	0.233
60-min trend	0.194
15-min trend	0.193
Personal time above range (learned)	0.168
2 h glucose variability	0.164

Time of day ranks high for highs because this is a hospital cohort with fixed meal times. In outpatients we expect logged meals to take that role.

Mean absolute tree-SHAP contribution in log-odds on held-out predictions (full twin). The dashboard shows the same values per alert.

LIMITATIONS, STATED PLAINLY

What this proof of concept does not show yet

One cohort: Chinese adults, mostly monitored in hospital. Indian diets, outpatient life and Indian CGM users need their own validation.

Short memory: 3 to 14 days per period, so the twin learns habits from at most two weeks.

No heart rate, steps or sleep in this dataset. The wearable adapters are designed but untested, so we make no claim about them.

Meal size comes from logged grams of food, not carbohydrate counts.

Retrospective only. No prospective or silent-mode trial yet, and not a medical device.

Associations, not causes: the twin warns and explains, but it does not simulate the effect of a dose change.

NEXT STEPS

Path to an Indian deployment

Step	What	Success measure
Wearables	Add HR, HRV, steps and sleep from Apple Health / Google Fit as dynamic features	Measure the AUROC gain
Records	Read EHR through FHIR R4 / ABDM instead of a spreadsheet	Same twin, live input
Indian cohort	Retrain and recalibrate on Indian CGM data with consent under DPDP	Calibration per site
Silent trial	Run beside a ward for 4 weeks; compare alerts with what happened	Alert burden, lead time
Mechanistic core	Pair with a physiological glucose model for what-if dose simulation	True counterfactuals

Summary

Real data: 100 patients with type 2 diabetes, EHR plus 15-min CGM, fused into one twin state.

Held-out patients: AUROC 0.878 for highs and 0.907 for lows two hours ahead, clearly above a trend rule.

Measured, not assumed: the twin's memory of the patient adds accuracy for highs; the static record, in this cohort, does not.

A clinician dashboard that replays the twin and explains every alert.

Open source under MIT. One command reproduces every number.

---

github.com/chxmq/SnackOverflow_RVCE · chxmq.github.io/SnackOverflow_RVCE

Team Snack Overflow · RV College of Engineering